QQ-W-461g
March 15, 1968
SUPERSEDING
Fed. Spec. QQ-W-461f
June 12, 1964

FEDERAL SPECIFICATION

WIRE, STEEL, CARBON (ROUND, BARE, AND COATED)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers round carbon steel wire, bare and coated.

1.2 Classification.

1.2.1 Compositions. The steel wire shall be of the following compositions, as specified (see 6.2), in accordance with Fed. Std. No. 66:

AISI number	
1006	1018
1008	1020
1010	1035
1015	1045

1.2.2 Finishes. The wire shall be supplied in the following finishes as specified (see 6.2):

Finish 1 - Bare.
Finish 2 - Coppered.
Finish 3 - Tinned.
Finish 4 - Liquor.
Finish 5 - Zinc-coated.

1.2.3 Tensile strength. The wire shall be furnished in accordance with the following tensile strength designations (see table Ia and Ib, and 6.3), as specified (see 6.2):

Finish	Tensile strength designation
Nos. 1, 2, 3, and 4	Hard, annealed in process, or annealed.
No. 5	Hard, medium, or soft.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

1.2.4 Weight of zinc-coating (finish 5). Zinc-coated wire shall be furnished in the following weights as specified (see 6.2):

Class 1 - Light.
Class 2 - Medium.
Class 3 - Heavy.

1.2.5 Wire size. Wire sizes shall be as specified in the contract or order (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

QQ-S-781 - Steel, Strapping, Flat.
QQ-S-790 - Steel Strapping, Round (Bare and Zinc-Coated).

Federal Standards:

Fed. Std. No. 48 - Tolerances for Steel and Iron Wrought Products.
Fed. Std. No. 66 - Steel: Chemical Composition and Hardenability.
Fed. Std. No. 123- Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specification required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-163 - Steel Mill Products Preparation for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply:

National Classification Board:

National Motor Freight Classification Rules.

(Application for copies should be addressed to the National Classification Board., 1616 P St., N. W., Washington, D. C. 20036.)

Uniform Classification Committee:

Uniform Freight Classification Rules.

(Application for copies should be addressed to Uniform Classification Committee, 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Material. The wire shall be carbon steel of the specified composition.

3.2 Practice. Unless otherwise specified (see 6.2), steel-making practice (i.e., rimmed, capped, semi-killed or killed steel) shall be at the manufacturer's option.

3.3 Chemical requirements.

3.3.1 Ladle analysis. A ladle analysis of each heat shall be furnished by the supplier.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

3.3.2 Check analysis. When specified (see 6.2), a check analysis shall be performed on the wire by the purchaser. A check analysis is not technologically appropriate on rimmed or capped steels. When the purchaser or contracting officer specifies that a check analysis is to be performed, the wire should not be made of rimmed or capped steels and the manufacturer or supplier should consider this in submission of bids.

3.3.2.1 The check analysis shall meet the requirements for the designated steel number within the tolerances indicated in Fed. Std. No. 66.

3.4 Dimensional requirements of wire.

3.4.1 Bare, coppered, tinned, and liquor finish wire. The diameter (size) of bare, coppered, tinned, or liquor finish wire shall not vary by more than the amount indicated in table 17a1 of Fed. Std. No. 48 for the size of wire specified.

3.4.2 Zinc-coated wire. The diameter (size) of zinc-coated wire shall not vary by more than the amount indicated in table 17a2 of Fed. Std. No. 48 for the size of wire specified.

3.4.3 For wire sizes less than 0.035 inch, the diameter of wire shall not vary by more than the amount indicated in tables 17a3, 17a4, 17a5, as applicable, of Fed. Std. No. 48 for the size of wire specified.

3.5 Tensile strength. The tensile strength shall conform with the requirements of table Ia and Ib, as applicable (see 1.2.3). Tensile ranges for finish 5 in AISI Nos. 1035 and 1045 shall be in accordance with the agreement between the purchaser and the manufacturer.

3.5.1 Elongation. Annealed wire in AISI numbers 1006, 1008, 1010, and 1015, for finishes 1 and 2, shall have an elongation not less than 15 percent in lengths of 10 inches. Annealed wire in AISI numbers 1018, 1020, 1035, and 1045, for finishes 1 and 2, shall have an elongation not less than 10 percent in lengths of 10 inches.

3.5.2 Mandrel wrapping for steel ductility. Bare or coated wire shall withstand, without fracture of the steel wire, wrapping in a closed helix in the manner specified in 4.4.5. This is not to be construed as a requirement for coatings.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

TABLE Ia. *Tensile strength requirements*

Wire diameter	Tensile strength, pounds per square inch				
	Hard		Annealed in process		Annealed
	Finishes 1, 2, 3, 4		Finishes 1, 2, 3, 4		Finishes 1, 2, 3, 4
	Minimum	Maximum	Minimum	Maximum	Maximum
Inches	AISI Number 1006				
0.499—0.328	50,000	80,000	47,000	67,000	60,000
.327— .250	50,000	85,000	50,000	70,000	60,000
.249— .192	60,000	95,000	50,000	70,000	60,000
.191— .178	65,000	100,000	50,000	70,000	60,000
.177— .148	70,000	105,000	55,000	80,000	60,000
.147— .128	75,000	110,000	55,000	80,000	60,000
.127— .106	80,000	120,000	55,000	80,000	60,000
.105— .086	85,000	130,000	55,000	85,000	60,000
.085— .072	90,000	140,000	60,000	90,000	60,000
.071— .058	100,000	150,000	60,000	90,000	60,000
.057— .048	105,000	155,000	60,000	90,000	60,000
.047— .035	—	—	60,000	90,000	60,000
Less than .035	—	—	60,000	95,000	60,000
	AISI Numbers 1008 and 1010				
0.499—0.328	55,000	85,000	53,000	73,000	65,000
.327— .250	60,000	90,000	53,000	73,000	65,000
.249— .192	65,000	100,000	55,000	75,000	65,000
.191— .178	70,000	105,000	55,000	75,000	65,000
.177— .148	75,000	110,000	58,000	83,000	65,000
.147— .128	80,000	115,000	58,000	83,000	65,000
.127— .106	85,000	125,000	58,000	83,000	65,000
.105— .086	90,000	135,000	60,000	90,000	65,000
.085— .072	100,000	145,000	60,000	90,000	65,000
.071— .058	110,000	155,000	60,000	90,000	65,000
.057— .048	—	—	60,000	90,000	65,000
.047— .035	—	—	60,000	90,000	65,000
Less than .035	—	—	60,000	95,000	70,000
	AISI Number 1015				
0.499—0.328	60,000	90,000	55,000	75,000	70,000
.327— .250	65,000	95,000	55,000	75,000	70,000
.249— .192	70,000	105,000	58,000	78,000	70,000
.191— .178	75,000	115,000	58,000	78,000	70,000
.177— .148	80,000	120,000	60,000	85,000	70,000
.147— .128	85,000	125,000	60,000	85,000	70,000
.127— .106	90,000	130,000	60,000	85,000	70,000
.105— .086	95,000	140,000	65,000	95,000	70,000
.085— .072	105,000	155,000	65,000	95,000	70,000
.071— .058	—	—	65,000	95,000	70,000
.057— .048	—	—	65,000	95,000	70,000
.047— .035	—	—	65,000	95,000	70,000
Less than .035	—	—	65,000	95,000	75,000

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

TABLE Ia. *Tensile strength requirements (cont'd)*

Wire diameter	Tensile strength, pounds per square inch				
	Hard		Annealed in process		Annealed
	Finishes 1, 2, 3, 4		Finishes 1, 2, 3, 4		Finishes 1, 2, 3, 4
	Minimum	Maximum	Minimum	Maximum	Maximum
Inches	AISI Number 1018 and 1020				
0.499—0.328	65,000	100,000	60,000	80,000	75,000
.327— .250	70,000	105,000	60,000	80,000	75,000
.249— .192	75,000	115,000	62,000	82,000	75,000
.191— .178	80,000	120,000	62,000	82,000	75,000
.177— .148	85,000	125,000	65,000	90,000	75,000
.147— .128	95,000	140,000	65,000	90,000	75,000
.127— .106	105,000	150,000	65,000	90,000	75,000
.105— .086	115,000	165,000	70,000	100,000	75,000
.085— .072	—	—	70,000	100,000	75,000
.071— .058	—	—	70,000	100,000	75,000
.057— .048	—	—	70,000	100,000	75,000
.047— .035	—	—	70,000	100,000	75,000
Less than .035	—	—	70,000	100,000	75,000
	AISI Number 1035				
0.499—0.328	80,000	120,000	70,000	100,000	85,000
.327— .250	85,000	125,000	70,000	100,000	85,000
.249— .192	90,000	135,000	70,000	100,000	85,000
.191— .178	95,000	140,000	70,000	100,000	85,000
.177— .148	100,000	145,000	75,000	110,000	85,000
.147— .128	110,000	160,000	75,000	110,000	85,000
.127— .106	125,000	180,000	75,000	110,000	85,000
.105— .086	135,000	190,000	75,000	110,000	85,000
Less than .086	—	—	75,000	110,000	85,000
	AISI Number 1045				
0.499—0.328	100,000	140,000	80,000	110,000	100,000
.327— .250	100,000	140,000	80,000	110,000	100,000
.249— .192	105,000	150,000	80,000	110,000	100,000
.191— .178	115,000	160,000	85,000	115,000	100,000
.177— .148	125,000	175,000	85,000	120,000	100,000
.147— .128	135,000	190,000	90,000	125,000	100,000
.127— .106	145,000	205,000	90,000	125,000	100,000
.105— .086	160,000	220,000	90,000	125,000	100,000
Less than .086	—	—	90,000	125,000	100,000

Table Ib. Tensile strength requirements for finish 5 (zinc-coated) (AISI numbers, 1006, 1008, 1010, 1018 and 1020)

Pounds per square inch

Size (inch)	Soft (steel numbers 1006, 1008, and 1010 only)	Medium	Hard
Over 0.176	70,000 (max.)	60,000— 85,000	75,000—105,000
0.176 to 0.106	70,000 (max.)	65,000— 90,000	80,000—110,000
0.105 to 0.080	75,000 (max.)	70,000— 95,000	85,000—115,000
0.079 to 0.035	75,000 (max.)	70,000—100,000	90,000—120,000
Less than 0.035	90,000 (max.)	75,000—110,000	100,000—130,000

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

3.6 Finish.

3.6.1 Bare wire. Bare wire shall be free from rust. Hard and annealed-in-process wire shall have a characteristic finish of either a grayish cast or slightly brownish cast (i.e., natural finish resulting from the dry method of drawing wire), at the option of the contractor. The annealed wire shall have a black finish or a grayish (lime bright annealed) finish as specified (see 6.2). If neither finish is specified, either the black finish or the grayish (lime bright annealed) finish may be furnished, at the option of the contractor. Black annealed-finish wire shall be free from excessive oxide scale.

3.6.2 Coppered wire. Coppered wire shall have a coating of copper applied by electroplating or be chemically deposited by immersion in a copper-sulphate solution or solutions before the wire is drawn through a die.

3.6.3 Tinned wire. Tinned wire shall have a coating of tin applied by the hot dip (molten bath) or electroplating process. The weight of tin per square foot of uncoated surface for the size of wire specified shall conform to the requirements of table II.

TABLE II. Weight of tin coating

Diameter of wire, inch	Weight of tin coating minimum, oz./sq. ft.
0.035 to 0.0625	0.015
.0626 to .1205	.020
.1206 to .1620	.025
.1621 to .1920	.030
.1921 and over	.035

3.6.4 Zinc-coated wire.

3.6.4.1 Zinc coating. Zinc-coated (galvanized) wire shall be uniformly and completely coated with zinc applied by either the hot-dip (molten-bath) or electroplating process. When the hot-dip process is employed the slab zinc used shall contain not less than 98 percent pure zinc.

3.6.4.2 Weight of zinc coating. The weight of zinc per square foot of uncoated wire surface shall comply with the requirements of table III, for the type of zinc coating specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

TABLE III. Zinc-coated wire: weight of zinc coating

Diameter		Class 1 coating	Class 2 coating	Class 3 coating
Steel wire Gage number	Inch	Wt. of zinc, oz./sq.ft., (min.)	Wt. of zinc, oz./sq.ft., (min.)	Wt. of zinc, oz./sq.ft., (min.)
5 and coarser	0.207 and coarser	0.65	0.75	0.90
6	.192	.50	.70	.90
7	.177	.40	.60	.80
8	.162	.40	.60	.80
9	.148	.40	.60	.80
10	.135	.30	.50	.80
11	.121	.30	.50	.80
12	.106	.30	.50	.80
12-1/2	.099	.30	.50	.80
13	.092	.30	.50	.70
14	.080	.25	.45	.65
14-1/2	.076	.20	.40	.60
15	.072	.15	.35	.50
16	.063	.15	.35	.50
17	.054	.15	.30	.40
18	.0475	.15	.30	.40
19	.0410	.10	.30	.40
20	.0348	.10	.30	.40
--	.0347 to 0.015	.07	--	--

Note: Fractional gages, other than those shown above, are produced with zinc-coatings equivalent to those of the next smaller size.

3.6.5 Adherence of coating. Copper, tin, liquor and zinc coating shall not flake or be removable by rubbing with the bare fingers when the wire is subject to the test specified in 4.4.6 and 4.4.6.1.

3.6.6 Liquor finish. Liquor finish wire may be supplied annealed, annealed-in-process, or hard. When supplied as annealed it shall be bright annealed.

3.7 Workmanship.

3.7.1 The wire shall be of uniform quality, clean, free from kinks, waviness, splits, cracks, and from injurious laps, seams, scale, segregation,

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

decarburization, and other injurious defects detrimental to the forming, machining, or fabrication of finished parts. Kinks shall be held to a minimum on annealed wire as agreed upon between the supplier and the purchaser.

3.7.2 Wire shall be furnished in coils, spools or reels without joints of any kind, except for welds made prior to drawing on hard and annealed-in-process, welds made prior to galvanizing and tinning, and welds made at finished size on annealed wire.

3.7.3 Coppered, tinned, zinc-coated and liquor finish wire shall be commercially free from bare or imperfectly coated spots, and shall be smooth, uniform, and scratch free.

3.8 Marking. Unless otherwise specified (see 6.2), one identification tag shall be attached to each coil, spool, or reel. Identification tags shall be marked with the name of the material, steel number, finish number, tensile strength designation, weight of coating designation, nominal size or gage, and number of contract or order.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot size. Unless otherwise specified (see 6.2), a lot shall consist of all wire submitted for inspection at the same time, of the same composition, tensile strength, designation, size, finish, and, when applicable, coating weight, and manufactured by the same process from steel of the same composition and heat.

4.3 Sampling.

4.3.1 Sampling for check analysis. For check analysis (see 3.3.2), wire shall be taken from each lot to prepare a sample weighing not less than 2 ounces. The sample shall consist of fine chips of metal, free of oil, and other foreign matter that would interfere with a chemical analysis.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

4.3.2 Sampling for examination and mechanical and coating tests. Seven sample units or ten percent of the units, whichever is less, shall be randomly selected from each lot. The inspection unit shall be one coil, spool, or reel of wire.

4.3.3 Test samples. A test sample of sufficient length shall be taken from each sample unit. Additional lengths may be taken to determine compliance with any special requirements contained in the contract or order. Each test sample taken shall be identified by sample unit and lot number.

4.4 Inspection and tests. Unless otherwise specified (see 6.2), inspection and tests shall be conducted in accordance with the applicable methods described in Fed. Test Method Std. No. 151.

4.4.1 Ladle analysis. Ladle analysis (see 3.3.1) shall be in accordance with the manufacturer's standard practice.

4.4.2 Check analysis. When tested in accordance with method 111.1 of Fed. Test Method Std. No. 151, the check analysis shall be in accordance with the requirements of 3.3.2.

4.4.3 Tensile strength. One test specimen, free from kinks and not less than 15 inches long, shall be taken from each test sample. The distance between the jaws of the testing machine, with the specimen ready for testing, shall be not less than 10 inches. When tested in accordance with method 211.1 of Fed. Test Method Std. No. 151, the tensile strength of the wire shall conform to the requirements of 3.5 and tables Ia and Ib, as applicable.

4.4.4 Elongation. For annealed wire, the elongation shall be determined during the tensile-strength test as the permanent increase in length after failure of a marked section originally 10 inches in length. The elongation shall comply with the requirements of 3.5.1.

4.4.5 Mandrel wrapping for steel ductility. One test specimen shall be taken from each test sample. The specimen shall be not less than 12 inches long and shall not be crooked or kinked. The specimen shall be wrapped in a closed helix for at least six complete turns around a cylindrical mandrel. For wire 0.1620 inch in diameter and smaller, the diameter of the mandrel shall be equal to the diameter of the wire; for wire larger than 0.1620 inch up to

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

and including 0.3120 inch in diameter, the mandrel diameter shall be twice the diameter of the wire. The wrapping test for wire larger than 0.3120 inch in diameter shall not be required, unless specified and only in accordance with test conditions agreed to between the supplier and the purchaser. Direct visual examination without optical magnification after wrapping shall be made for compliance with the requirements of 3.5.2.

4.4.6 Adherence of copper, tin, and liquor coatings. One test specimen, not less than 12 inches long and not crooked or kinked, shall be taken from each test sample. The specimen shall be wrapped at a rate of not more than 15 turns per minute, in a closed helix of at least 2 turns around a cylindrical mandrel. The mandrel shall have a diameter equal to 3 times the nominal diameter of the wire specimen for AISI numbers 1006, 1008, 1010, and 1015, and 5 times the nominal diameter of the wire specimen for AISI numbers 1018, 1020, 1035, and 1045. The wire surface of the helix shall be examined for compliance with the requirements of 3.6.5. Visual examination shall be made without optical magnification.

4.4.6.1 Adherence of zinc coating for classes 1, 2, and 3. For zinc-coated wire, one test specimen, not less than 12 inches long and not crooked or kinked, shall be taken from each test sample. The specimen shall be wrapped at a rate of not more than 15 turns per minute in a helix of at least 8 turns, with a pitch approximately equal to the diameter of the wire, around a cylindrical mandrel. The diameter of the mandrel shall be equal to three times the nominal diameter of the wire. The wire surface of the helix shall be examined for compliance with the requirements of 3.6.5. Visual examination shall be made without optical magnification.

4.4.7 Weight of tin or zinc coating. One test specimen not less than 12 inches long shall be cut from each test sample. The computed weight of coating for each test specimen shall be not less than the minimum specified. The test for zinc or tin coating shall be conducted as outlined in stripping test of method 512.1 of Fed. Test Method Std. No. 151. (While the title for method 512.1 indicates that the method is for zinc, the method can also be used for tin.)

4.4.8 Zinc coating (hot dip). When a chemical analysis test is conducted in conjunction with the hot dip (molten bath) process to determine compliance with 3.6.4.1, the slab zinc samples shall contain not less than 98 percent zinc. If the samples tested contain less than 98 percent pure zinc the entire quantity of wire processed

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

in the bath shall be rejected. As an alternate to analyzing the metal in the galvanizing kettle at the time the wire furnished to this specification is coated, the Government will accept certification from the producer that the coating metal used was of Prime Western or better chemical composition.

4.4.9 Dimensions. Sample units shall be selected in accordance with 4.3.2 for dimensional inspection. Each unit selected shall be gaged at 3 places, if accessible, one near each end and one near the middle. If not accessible, the wire from each unit shall be gaged by unreeling 12 feet from each unit and gaging in 3 places between the second and twelfth foot from the end. A unit shall be considered nonconforming if any measurement is not within specified limits, and shall be considered cause for rejection.

4.5 Inspection of preparation for delivery. The preservation, packaging, packing, and marking of the wire shall be examined to determine compliance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as required (see 6.2).

5.1.1 Level A. The wire shall be preserved and packaged in accordance with the applicable level A requirements of MIL-STD-163, with the following additional requirements.

5.1.1.1 Catchweight coils. Unless otherwise specified (see 6.2), the wire shall be delivered in catchweight (one piece) coils. Each catchweight coil shall be secured with not less than three double bands of zinc-coated wire or three single bands of steel strapping conforming to QQ-S-790 (finish 2), and QQ-S-781 (type 1, class B).

5.1.1.2 Exact-weight coils, spools, or reels. When exact-weight coils, spools, or reels are ordered, the inside and outside diameters of the coils, spools, or reels shall be specified (see 6.2), as required.

5.1.1.3 When specified, AISI numbers 1006, 1008, and 1010 of annealed wire, 0.023 to 0.063 inch in diameter, inclusive, shall be furnished in 5- and 12-pound even-weight coils, or in 5-pound spools. Coils

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

and spools shall be paper-wrapped between flanges. Twelve-pound coils (stones) shall have a mean diameter of approximately 8 inches and unless otherwise specified (see 6.2), there shall be not more than two pieces per coil. Unless otherwise specified (see 6.2), 12-pound coils shall be packaged in barrels or fiber drums of 42 coils. Three 5-pound coils may be wrapped as a unit, at the supplier's option.

5.1.2 Level C. The wire shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 Packing. Packing shall be level A, B, or C as required (see 6.2).

5.2.1 Level A. The wire shall be packed in accordance with the applicable level A requirements of MIL-STD-163.

5.2.2 Level B (see 6.4). The wire shall be packed in accordance with the applicable level B requirements of MIL-STD-163.

5.2.3 Level C. The materials shall be packed in a manner which will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.3 Marking, (see 6.2).

5.3.1 Civil agencies. In addition to any special marking specified in the contract or order, the wire shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking specified in the contract or order, the wire shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This specification covers the more generally available grades of carbon-steel wire for use where specific compositions and properties are desired for a definite purpose. It is not intended that this specification apply to items such as rope wire, armor wire, etc., which are covered by individual Federal Specifications.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. Purchasers should select the preferred options herin and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Composition (see 1.2.1).

(c) Finish (see 1.2.2).

(d) Tensile strength designation (see 1.2.3).

(e) Weight of zinc-coating, if required (see 1.2.4).

(f) Size of wire (see 1.2.5).

(g) Type of steel if different (see 3.2).

(h) Check analysis, if required (see 3.3.2).

(i) Finish required (see 3.6.1).

(j) Marking (see 3.8 and 5.3).

(k) Lot composition if different (see 4.2).

(1) Inspection and tests if different (see 4.4).

(m) Level of preservation and packaging and level of packing required (see 5.1 and 5.2).

(n) Weight of coils, spools, or reels, if required (see 5.1.1.1).

(o) Size of exact-weight coils, spools, or reels, if required (see 5.1.1.2).

6.2.1 Information on end use of the wire shall be furnished whenever possible to help suppliers furnish material best suited for the intended purpose.

6.3 Hard wire in grades 1020, 1035, and 1045 as covered in this specification refers to wire drawn to finished size from heat-treated rods or produced by other special mill procedure. Annealed wire refers to wire annealed at finished size and annealed-in-process

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.

refers to wire annealed during manufacturing operations. In the production of soft, medium or hard zinc-coated wire, the heat treatment which determines the final strength is performed in conjunction with and as part of the galvanizing operation.

6.4 For civil agency procurements. When level B packaging and packing is required for civil agency procurements, the requirements for level C shall apply.

MILITARY INTERESTS:

Custodians:

Army - MR
Navy - YD
Air Force - 84

Review activities:

Army - MU, WC, MI, MR, AT
Navy - YD, MC
Air Force - 84, 85

User activity:

Army - AV, ME

Preparing activity:

Navy - YD

Order for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents See section 2 of this specification to obtain extra copies and other documents referenced herein. cents each.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:06Z
Check the source to verify that this is the current version before use.